13 The number of animals in a population at the start of year t is P_t
The number of animals at the start of year 1 is 400

Given that

$$P_{t+1} = 1.01P_t$$

work out the number of animals at the start of year 3

$P_1 = 400$

$P_2 = 1.01 \times 400 = 404$

$P_3 = 1.01 \times 404 = 408.04$

1 mark
Multiply by 1.01
for each next year.

1 mark
Year 3 needs two steps from year 1.

408

(Total for Question 13 is 2 marks)

14 y is inversely proportional to x^3

$y = 44$ when $x = a$

Show that $y = 5.5$ when $x = 2a$

(Total for Question 14 is 3 marks)

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